

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location Google: Arcola 2, VA -- station KIAD, America/New_York
Committed pair OSM 693381107 -> 545396372
Amazon IAD81 / Amazon IAD62
Facade gap 137.7 m between the two halls
Weather record 43,763 real hourly records from KIAD
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3539 C at 225 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2025-03-11 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 2 of 24 hours with 1 mode change. The reactive incumbent took 13 hours -- 11 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 2 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 13 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	4.329	18.0	4.132	1.053
01	mechanical	yes	switch budget	4.329	18.0	4.132	0.994
02	mechanical	yes	switch budget	4.049	18.0	3.022	1.163
03	mechanical	yes	switch budget	2.918	18.0	1.183	1.018
04	mechanical	yes	switch budget	2.944	18.0	0.802	0.992

05	mechanical	yes	switch budget	2.359	18.0	0.242	1.168
06	mechanical	yes	switch budget	0.999	18.0	-0.868	0.983
07	mechanical	yes	switch budget	2.664	18.0	0.242	1.636
08	mechanical	yes	switch budget	6.231	18.0	5.579	1.912
09	mechanical	yes	switch budget	9.974	18.0	11.442	2.358
10	mechanical	yes	switch budget	11.931	18.0	14.216	2.254
11	mechanical	yes	switch budget	15.598	18.0	17.799	1.798
12	mechanical	no	dry-bulb	18.603	18.0	19.564	1.367
13	mechanical	no	dry-bulb	19.906	18.0	21.364	1.173
14	mechanical	no	dry-bulb	22.220	18.0	22.023	1.293
15	mechanical	no	dry-bulb	22.629	18.0	23.103	1.070
16	mechanical	no	dry-bulb	23.301	18.0	23.103	1.143
17	mechanical	no	dry-bulb	22.874	18.0	22.543	1.416
18	mechanical	no	dry-bulb	22.479	18.0	21.219	1.328
19	mechanical	no	dry-bulb	20.946	18.0	19.014	1.418
20	mechanical	no	dry-bulb	20.161	18.0	17.799	1.558
21	mechanical	no	dry-bulb	18.825	18.0	15.859	1.691
22	FREE	yes	-	16.065	18.0	12.473	1.374
23	FREE	yes	-	15.815	18.0	13.362	1.137

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.329 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.329 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.049 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.918 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.944 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a

thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.359 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.999 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.664 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.231 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.974 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.931 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.598 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.603 C against a 18.0 C limit, so it fails by 0.603 C. It would take a limit 0.603 C higher, or a bound 0.603 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.906 C against a 18.0 C limit, so it fails by 1.906 C. It would take a limit 1.906 C higher, or a bound 1.906 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.220 C against a 18.0 C limit, so it fails by 4.220 C. It would take a limit 4.220 C higher, or a bound 4.220 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.629 C against a 18.0 C limit, so it fails by 4.629 C. It would take a limit 4.629 C higher, or a bound 4.629 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.301 C against a 18.0 C limit, so it fails by 5.301 C. It would take a limit 5.301 C higher, or a bound 5.301 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.874 C against a 18.0 C limit, so it fails by 4.874 C. It would take a limit 4.874 C higher, or a bound 4.874 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.479 C against a 18.0 C limit, so it fails by 4.479 C. It would take a limit 4.479 C higher, or a bound 4.479 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.946 C against a 18.0 C limit, so it fails by 2.946 C. It would take a limit 2.946 C higher, or a bound 2.946 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.161 C against a 18.0 C limit, so it fails by 2.161 C. It would take a limit 2.161 C higher, or a bound 2.161 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.825 C against a 18.0 C limit, so it fails by 0.825 C. It would take a limit 0.825 C higher, or a bound 0.825 C tighter, to change this hour.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.065 C, which is 1.935 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.374 C, and the margin is measured, not chosen: 1.189 C of group-conditional forecast error for this hour of day, plus 0.1849 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.815 C, which is 2.185 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.137 C, and the margin is measured, not chosen: 0.953 C of group-conditional forecast error for this hour of day, plus 0.1841 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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